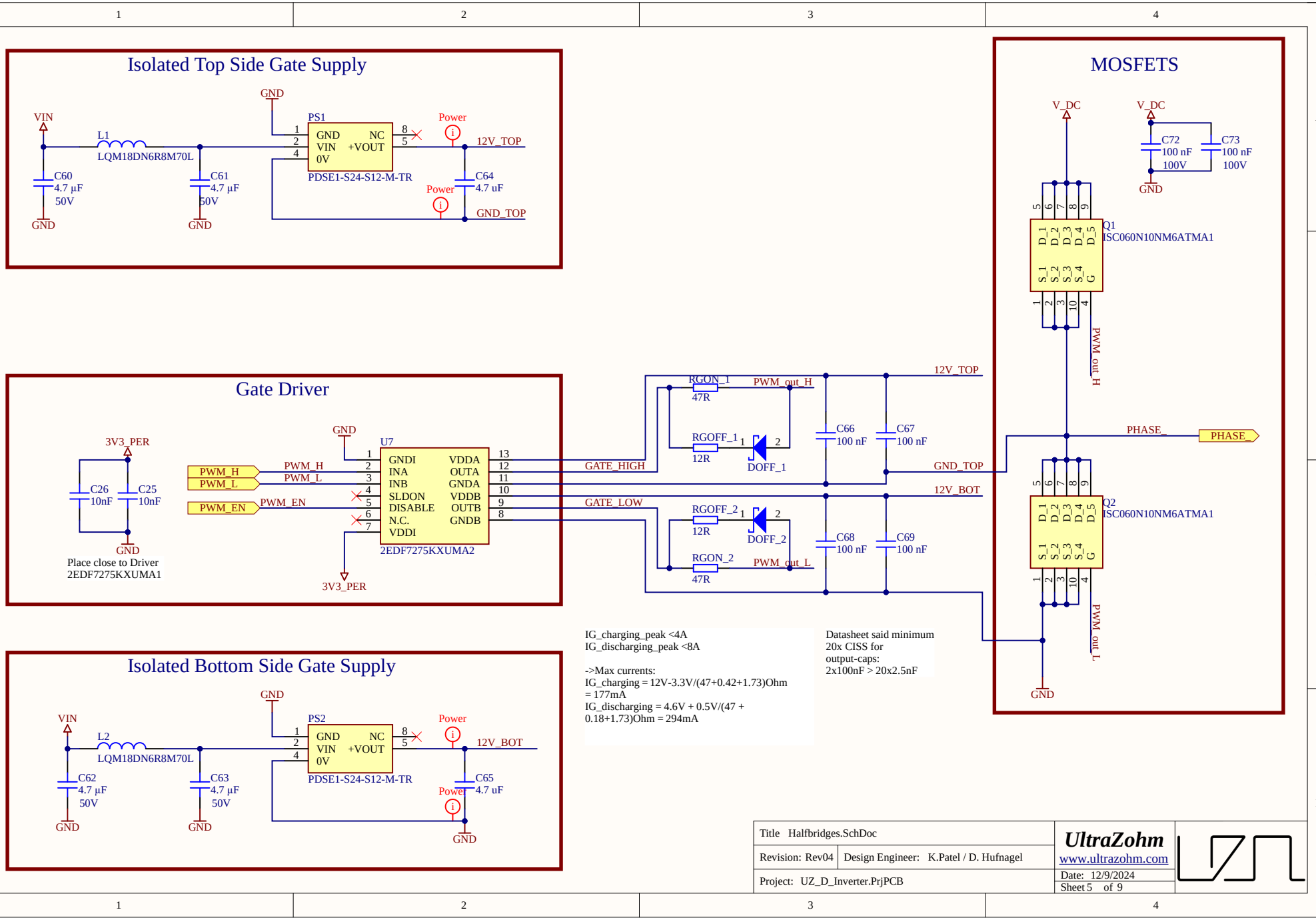
Isolated Top Side Gate Supply
VIN
L1
LQM18DN6R8M70L
C60
4.7 µF
50V
C61
4.7 µF
50V
GND
PS1
GND
VIN
0V
NC
+VOUT
PDSE1-S24-S12-M-TR
Power
12V_TOP
C64
4.7 uF
GND_TOP
Gate Driver
3V3_PER
C26
10nF
C25
10nF
Place close to Driver
2EDF7275KXUMA1
PWM_H
PWM_L
PWM_EN
U7
GNDI
INA
INB
SLDON
DISABLE
N.C.
VDDI
VDDA
OUTA
GNDA
VDDB
OUTB
GNDB
2EDF7275KXUMA2
GATE_HIGH
GATE_LOW
RGON_1
47R
RGOFF_1
12R
DOFF_1
RGOFF_2
DOFF_2
RGON_2
C66
100 nF
C67
100 nF
C68
C69
Isolated Bottom Side Gate Supply
L2
C62
C63
C65
PS2
12V_BOT
MOSFETS
V_DC
C72
C73
100 nF
100V
Q1
Q2
ISC060N10NM6ATMA1
PWM_out_H
PWM_out_L
PHASE_
IG_charging_peak <4A
IG_discharging_peak <8A
->Max currents:
IG_charging = 12V-3.3V/(47+0.42+1.73)Ohm
= 177mA
IG_discharging = 4.6V + 0.5V/(47 +
0.18+1.73)Ohm = 294mA
Datasheet said minimum
20x CISS for
output-caps:
2x100nF > 20x2.5nF
Title Halfbridges.SchDoc
Revision: Rev04
Design Engineer: K.Patel / D. Hufnagel
Project: UZ_D_Inverter.PrjPCB
UltraZohm
www.ultrazohm.com
Date: 12/9/2024
Sheet 5 of 9